

2/4/19

How to Construct / guarantee that an estimator $w(x)$ is Best Unbiased —

(check technical conditions for CRLB)

if $\text{Var}_\theta[w] = \text{CRLB}(\theta)$
then w is "BUE"

(what if CRLB is not applicable)?

(Lehmann - Scheffe) theorem:
An alternative way to guarantee that an estimator is B.U.

Rao - Blackwell - Kolmogorov Theorem (1945-47):

Suppose $X \sim P_\theta$, $\theta \in \Theta$ and let
 $w(x)$ be an unbiased estimator of $g(\theta)$. let $T(x)$
be a sufficient statistic for θ . $\tilde{w}(x)$

Define the estimator $E_\theta[w(x)|T(x)] =: \tilde{\phi}(T(x))$

↑
does not depend on θ
as T sufficient

Then,

$$(i) \quad E_\theta[\tilde{\phi}(T(x))] = g(\theta), \forall \theta$$

$$(ii) \quad \text{Var}_\theta[\tilde{\phi}(T(x))] \leq \text{Var}_\theta[w(x)]$$

Proof:

$$\begin{aligned} (1) \quad g(\theta) &= E_\theta[w(x)] \\ &= E_\theta[E_\theta[w(x)|T(x)]] = E_\theta[\tilde{\phi}(T(x))] \end{aligned}$$

$X \sim P_\theta$

$$E_\theta[w(x)] = g(\theta)$$

$$E_\theta[\tilde{\phi}(T(x))] = g(\theta)$$

② lemma: for rvs X and Y

$$\text{Var}(X) = \text{Var}(E(X|Y)) + E[\text{Var}(X|Y)] \quad (68)$$

Proof of lemma:

$$\begin{aligned} \text{Var}(X) &= E[(X - E(X))^2] \\ &= E[E[(X - E(X))^2 | Y]] \\ &= E\left[E\left[\underbrace{(X - E(X|Y))}_A + \underbrace{E(X|Y) - E(X)}_B\right)^2 | Y\right]\right] \\ &= E[E(X - E(X|Y))^2] \end{aligned}$$

$$\text{Hence, } \text{Var}_\theta[w(X)] = \underbrace{\text{Var}_\theta(E_\theta(w(X)|T(X)))}_{\text{Var}_\theta[\tilde{w}(X)]} + \underbrace{E_\theta[\text{Var}_\theta(w(X)|T(X))]}_{\geq 0}$$

example $X = (X_1, \dots, X_n) \stackrel{\text{iid}}{\sim} \text{Poi}(d)$, $d \geq 0$ to be estimated

(consider the estimator $w(X) = X_1$,

$w(X)$ is unbiased for $g(d) = d$

$T(X) = \sum_{i=1}^n X_i$ is a sufficient
statistic for d

$$\tilde{w}(x) = E_d \left[x_1 \mid \sum_{i=1}^n x_i = s \right] = d(s) \text{ (say)}$$

$$h d(s) = \sum_{j=1}^n E_d \left[x_j \mid \sum_{i=1}^n x_i = s \right] \quad (\text{symmetry})$$

$$h d(s) = E_d \left[\sum_{j=1}^n x_j \mid \sum_{i=1}^n x_i = s \right] \\ = s$$

$$\boxed{d(s) = s/n} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\tilde{w}(x) = E_d \left[x_1 \mid \sum_{i=1}^n x_i \right] = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\Rightarrow d(s) = s/n$$

example: $X \equiv (x_1, \dots, x_n) \stackrel{\text{i.i.d.}}{\sim} \text{Unif}[0, \theta]$
 $\theta > 0$

$$w(x) = 2x_1$$

$$T(x) = \max_i x_i \text{ is suff. statistic for } \theta$$

$$\tilde{w}(x) = E_\theta [w(x) \mid T(x)] = E_\theta [2x_1 \mid \max_i x_i] \\ = \frac{(n+1)}{(n)} \max_i x_i \\ = d(s)$$

$$h d(s) = \sum_{j=1}^n E_\theta [2x_j \mid \max_i x_i = s] \\ = E_\theta \left[2 \sum_{j=1}^n x_j \mid \max_i x_i = s \right]$$

* Is the BUE Unique?

Theorem: If $w(x)$ is a BUE of $g(\theta)$, then it is the unique BUE of $g(\theta)$.

Proof

Suppose $w'(x)$ is another BUE of $g(\theta)$

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Consider

$w^*(x) = \frac{1}{2}(w(x) + w'(x))$, which is unbiased for $g(\theta)$.

$$\begin{aligned} \text{Var}_\theta(w^*) &= \frac{1}{4} \{ \text{Var}_\theta w + \text{Var}_\theta w' + 2 \cdot \text{Cov}_\theta(w, w') \} \\ &\leq \frac{1}{4} \{ 2 \text{Var}_\theta(w) + 2 \sqrt{\text{Var}_\theta(w) \cdot \text{Var}_\theta(w')} \} \\ &= \text{Var}_\theta(w) \end{aligned}$$

Since w is a BUE, equality must hold above

$$\Rightarrow w(x) - g(\theta) = \alpha(\theta) (w'(x) - g(\theta)) \quad \text{as } \frac{\partial}{\partial \theta}$$

$$w = \alpha(\theta) w' + \beta(\theta), \quad \forall \theta$$

$$\Rightarrow \text{Cov}_\theta(w, w') = \text{Cov}_\theta[\alpha(\theta) w' + \beta(\theta), w']$$

$$= \alpha^2(\theta) \text{Var}_\theta(w') + 0$$

$$= \text{Var}_\theta(w')$$

by Cauchy
Schwartz inequality

$$\Rightarrow \left. \begin{aligned} \alpha(\theta) &= 1 \quad \forall \theta \\ \beta(\theta) &= 0 \quad \forall \theta \end{aligned} \right\} w = w'$$

* Characterization of a BUE?

Theorem: Suppose $w(x)$ is an unbiased estimator for $g(\theta)$ when $X \sim P_\theta$, $\theta \in \Theta$. Then,

Then,

w is the best unbiased estimator of $g(\theta)$ iff $w(x)$ is uncorrelated with all unbiased estimators of 0 ($\forall \theta$)

Proof: Only if: let $w(x)$ be BUE of $g(\theta)$, and let $w'(x)$ be an unbiased estimator of 0 i.e.

$$E_\theta[w'(x)] = 0 \quad \forall \theta \in \Theta$$

Consider the estimator $w_a(x) = w(x) + a w'(x)$, $a \in \mathbb{R}$ which is unbiased for $g(\theta)$.

$$\begin{aligned} \text{Var}_\theta(w_a) &= \text{Var}_\theta(w + a w') \\ &= \text{Var}_\theta(w) + \underbrace{a^2 \text{Var}_\theta(w') + 2a \text{Cov}(w, w')}_{\star} \end{aligned}$$

if $\text{Cov}_\theta(w, w') < 0$

Can choose a

$$a \in \left(0, -2 \frac{\text{Cov}(w, w')}{\text{Var}(w')} \right)$$

to make $\star < 0$

$$\begin{aligned} &> 0 \\ &a(a \text{Var}(w') + 2 \text{Cov}(w, w')) \\ &\text{Var}(w) a \left(a + \frac{2 \text{Cov}(w, w')}{\text{Var}(w')} \right) \\ &a(a + c) \end{aligned}$$

if $\text{Cov}_\theta(w, w') > 0$

Can choose a to
make $\star < 0$.

Hence by the optimality of w ,

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$$\text{Cov}_\theta(w, w') = 0 \quad \forall \theta$$

✗ "of part": Assume $w(x)$ uncorrelated with all unbiased estimators of θ . Let $\tilde{w}(x)$ be an unbiased estimator of $g(\theta)$.

We will show $\text{Var}_\theta(w) \leq \text{Var}_\theta(\tilde{w})$

$$w'(x) = w(x) + (w'(x) - w(x))$$

$$\begin{aligned} \Rightarrow \text{Var}_\theta(w'(x)) &= \text{Var}_\theta[w(x) + (w'(x) - w(x))] \\ &= \text{Var}_\theta(w) + \underbrace{\text{Var}_\theta(w' - w)}_{\geq 0} + 2 \underbrace{\text{Cov}(w, w - w')}_{= 0} \\ &\geq \text{Var}_\theta(w) \end{aligned}$$

Definition: A family $\{P_\theta : \theta \in \Theta\}$ of pdf's / pmf is said to be COMPLETE if for any function g

$$\forall \theta \quad \mathbb{E}_\theta[g(x)] = 0 \implies \forall \theta \quad P_\theta[g(x) = 0] = 1$$

$X \sim P_\theta$

e.g. Consider the Binomial family: $\{ \text{Bin}(n, \theta), 0 \leq \theta \leq 1 \}$

This family is Complete i.e.

$$\forall \theta \in [0, 1]$$

$$E_0[g(T(x))] = 0$$

$$\sum_{i=0}^n g(i) \binom{n}{i} \theta^i (1-\theta)^{n-i} = 0$$

$$\Rightarrow \underline{\underline{g(0) = \dots = g(n) = 0}}$$

e.g. $\{N(\mu, 1) : \mu \in \mathbb{R}\}$
 $\{\text{unif}[0, \theta] : \theta > 0\}$

Defn: [Complete statistics] $\xrightarrow{\text{function/estimator of data}}$ A statistics

$T(x)$ is said to be Complete statistics for the family distribution $\{P_\theta : \theta \in \Theta\}$ if the family of distribution of $T(x)$ is a Complete family — $\left\{ \begin{array}{l} \forall \theta \quad E_\theta[g(T(x))] = 0 \\ \Rightarrow P_\theta(g(T(x)) = 0) = 1 \end{array} \right\}$

Theorem [Lehmann-Scheffe ≈ 1950]:

let $T(x)$ be a Complete, sufficient statistics when $X \sim P_\theta$, $\theta \in \Theta$; and let $\phi(T(x))$ be an estimator based on $T(x)$. Then $(\phi \circ T)$ is the best unbiased estimator of its own expected value.